List & dimensions

Suppose A is an (6)×(7) matrix with rank r = 4

A with r linearly independent rows in blue:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & a_{17} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & a_{26} & a_{27} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} & a_{36} & a_{37} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} & a_{46} & a_{47} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} & a_{56} & a_{57} \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} & a_{66} & a_{67} \end{bmatrix}$$

A with r linearly independent columns in blue:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & a_{17} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & a_{26} & a_{27} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} & a_{36} & a_{37} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} & a_{46} & a_{47} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} & a_{56} & a_{57} \\ a_{61} & a_{62} & a_{63} & a_{64} & a_{65} & a_{66} & a_{67} \end{bmatrix}$$

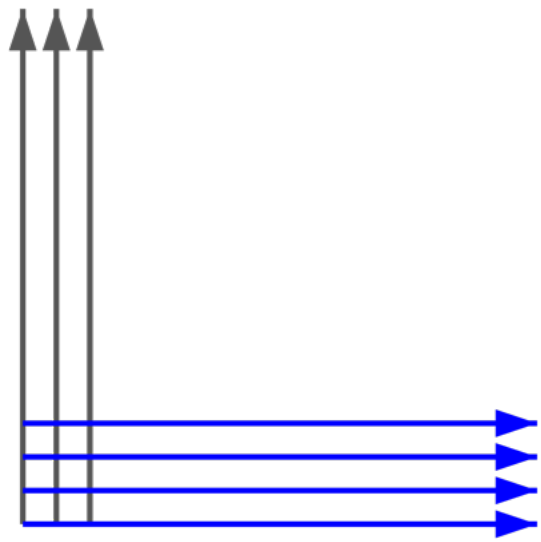
(note that gray and blue rows and columns can appear in any order)

Product $A \times \vec{x}$ is defined for all n×1 vectors $\vec{x}$

Domain of A is the set of all $\vec{x}$ where $A \times \vec{x}$ is defined or $\mathbb{R}^n$
Domain of A is spanned by two complementary subspaces:

① All linear combinations of independent rows of A
r-dimensional subspace of $\mathbb{R}^n$, called row space of A
(spanning directions of row space are shown as long blue arrows)

② Subset of all $\vec{x}$ where $A \times \vec{x} = \vec{0}$, called null space of A
(spanning directions of null space are shown as long gray arrows)



- $\mathbb{R}^n = \text{row}(A) \oplus \text{null}(A)$

↓

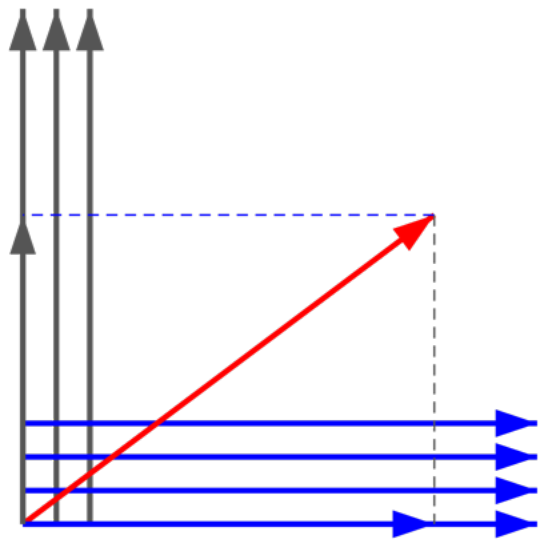
every $\vec{x} \in \mathbb{R}^n$ can be written uniquely as $\left(\vec{x}_{\parallel} \in \text{row}(A)\right) + \left(\vec{x}_{\perp} \in \text{null}(A)\right)$

- $\vec{x}$: red arrow
- $\vec{x}_{\parallel}$: short blue arrow
- $\vec{x}_{\perp}$: short gray arrow

- definition $A \times \vec{x} = \vec{0}$ implies that
every vector in null(A) is orthogonal to every row of A

↓

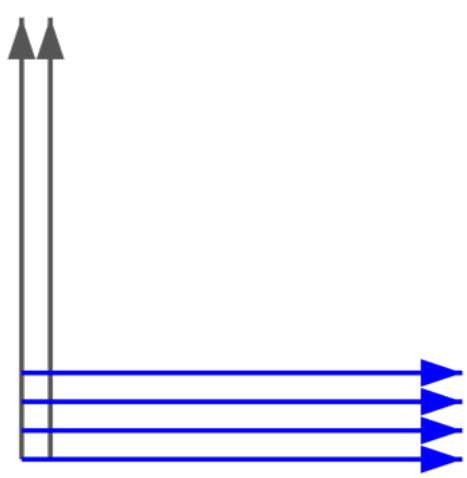
the complementary subspaces are orthogonal: $\text{row}(A) \perp \text{null}(A)$



Codomain of A is the set of all possible outputs $\vec{b} = A \times \vec{x}$ or $\mathbb{R}^m$
Codomain of A is spanned by two complementary subspaces:

① All linear combinations of independent columns of A
r-dimensional subspace of $\mathbb{R}^m$, called column space of A
(spanning directions of column space are shown as long blue arrows)

② Subset of all $\vec{b}$ orthogonal to every column of A, called left null space of A
same as subset of all $\vec{b}$ orthogonal to every row of A^T
computed as $A^T \vec{b} = \vec{0}$
(spanning directions of left null space are shown as long gray arrows)



• $\mathbb{R}^m = \text{col}(A) \oplus \ell\text{-null}(A)$

$\downarrow$

every $\vec{b} \in \mathbb{R}^m$ can be written uniquely as $\left(\vec{b}_{\parallel} \in \text{col}(A)\right) + \left(\vec{b}_{\perp} \in \ell\text{-null}(A)\right)$

- $\vec{b}$: red arrow
- $\vec{b}_{\parallel}$: short cyan arrow
- $\vec{b}_{\perp}$: short magenta arrow

• definition of left null space implies that
every vector in ℓ-null(A) is orthogonal to every column of A

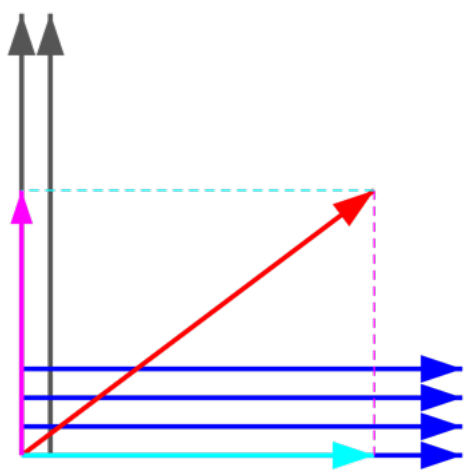
$\downarrow$

the complementary subspaces are orthogonal: $\text{col}(A) \perp \ell\text{-null}(A)$

• $\dim\left(\text{row}(A)\right) = \dim\left(\text{col}(A)\right) = r$

• $\dim\left(\text{null}(A)\right) = n - r$

• $\dim\left(\ell\text{-null}(A)\right) = m - r$



Four subspaces, codomain view

Suppose we have a 3×2 matrix A = [$\vec{a}_1$ | $\vec{a}_2$]
where $\vec{a}_1$ & $\vec{a}_2$ are linearly independent vectors in $\mathbb{R}^3$, rank(A) = 2

$$A = \begin{bmatrix} 0.8 & -0.9 \\ 1.1 & 1.2 \\ 0.6 & 0.5 \end{bmatrix}$$

① Codomain(A) is the set of all vectors in $\mathbb{R}^3$

② column space(A) or all linear combinations of $\vec{a}_1$ & $\vec{a}_2$
or the plane spanned by the two vectors in $\mathbb{R}^3$ and expressed in 3D coordinates
• Dimension: r=2 in $\mathbb{R}^3$
• Output does not fill the codomain
⇔
Transformation is not onto

③ ℓ-null space(A) or all directions in the codomain ($\mathbb{R}^3$) orthogonal to column space:
all vectors along the orthogonal line, shown in purple
• Dimension: m-r=1 in $\mathbb{R}^3$

④ Domain(A) = $\mathbb{R}^2$, an abstract 2D coefficient space, or an abstract plane

⑤ null space(A) or all directions in the domain ($\mathbb{R}^2$) that produce $\vec{0}$ output:
for the given A, only the $\vec{0}$ vector
• Dimension: n-r=0 in $\mathbb{R}^2$
• Every vector from the domain produces a unique output
⇔
Transformation is one-to-one

⑥ row space(A)

While rows of A cannot be directly visualized in the codomain image,
row space is the set of all vectors in the domain that are orthogonal to null space,
or, equivalently,
set of all vectors that have a 0 component in the null space
for the given A, all vectors in $\mathbb{R}^2$ satisfy the condition

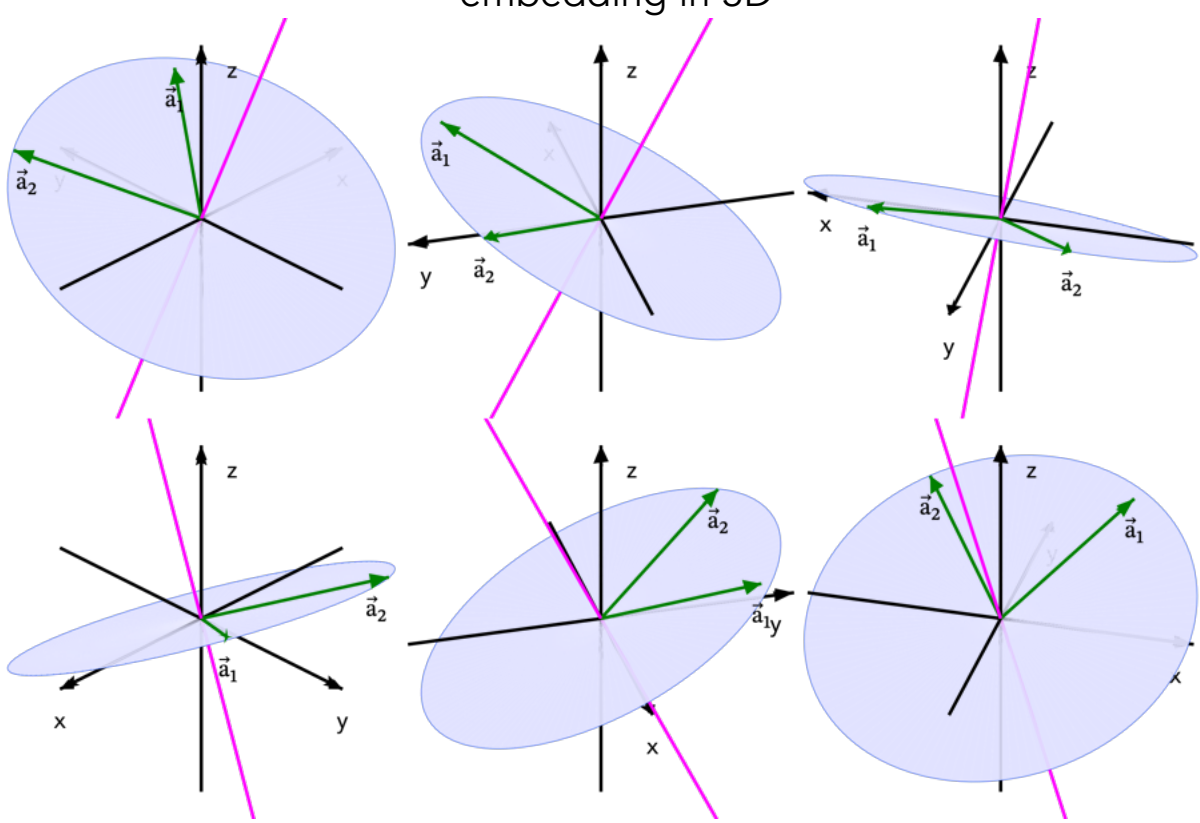
- Dimension: r=2 in $\mathbb{R}^2$

⑦ Matrix A takes a set of all vectors in $\mathbb{R}^2$ and transforms them in a way that

- (1, 0) maps to $\vec{a}_1$
- (0, 1) maps to $\vec{a}_2$
- all other vectors $\vec{x}$ follow as $x_1\vec{a}_1 + x_2\vec{a}_2$, which fills the blue plane in $\mathbb{R}^3$

Geometrically, A acts on a square 2D grid and results in

- combination of grid deformation and rotation
- embedding in 3D



Four subspaces, domain view

Suppose we have a 2×3 matrix B (unrelated to matrix A on the previous page) where $\vec{r}_1$ & $\vec{r}_2$ are linearly independent row vectors of B in $\mathbb{R}^3$, rank(B) = 2

$$B = [\, \vec{b}_1 \mid \vec{b}_2 \mid \vec{b}_3 \,] = \begin{bmatrix} \vec{r}_1 \\ \vec{r}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0.4 & 0.8 \\ 0.6 & 1.1 & 0.3 \end{bmatrix}$$

① Domain(B) = $\mathbb{R}^3$, an abstract 3D coefficient space

② row space(B)
Row space is the plane spanned by the two independent rows $\vec{r}_1$ and $\vec{r}_2$ in $\mathbb{R}^3$ and expressed in 3D coordinates

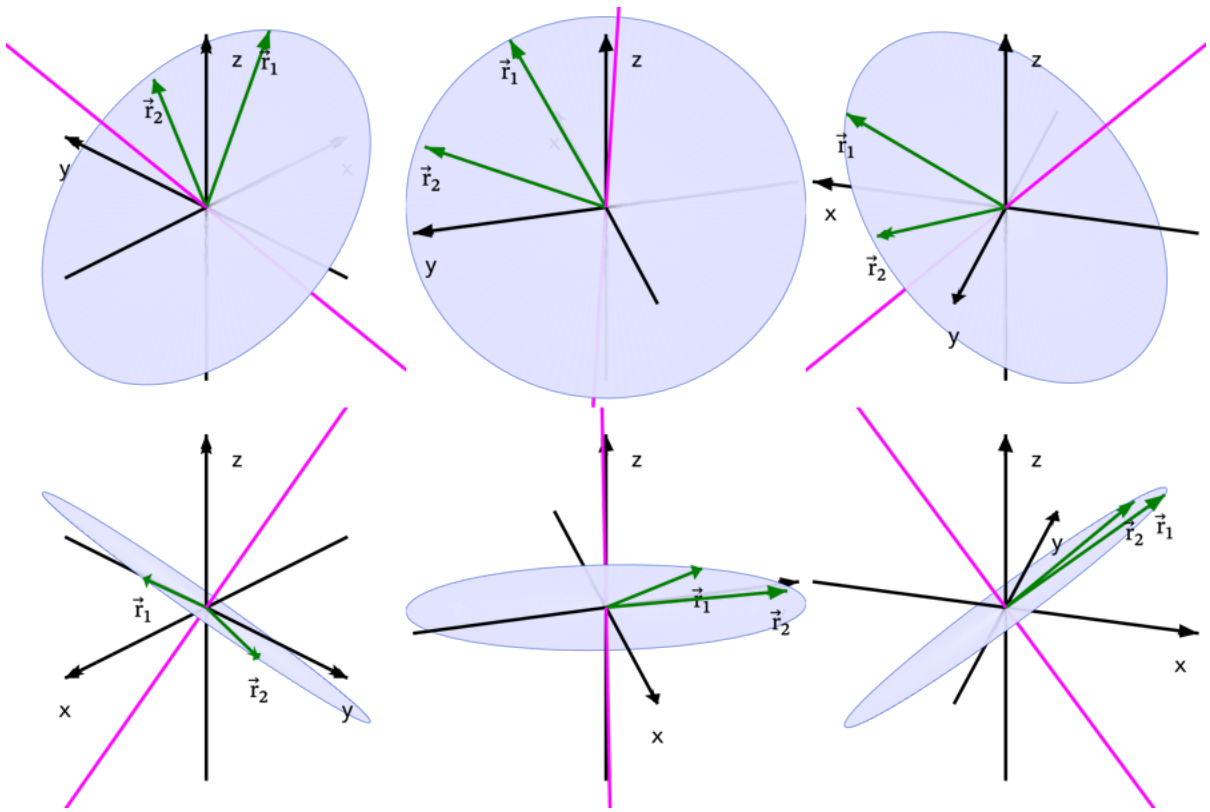
- Dimension: r=2 in $\mathbb{R}^3$

③ null space(B) or all directions in the domain ($\mathbb{R}^3$) that produce $\vec{0}$ output:
we can deduce it by solving B $\vec{x} = \vec{0}$
equivalently, $\vec{r}_i \cdot \vec{x} = 0$ for every i

Recall the 2D case: each row gave a constraint line through the origin
Here the same idea holds one dimension higher: each row gives a constraint plane

- Each equation $\vec{r}_i \cdot \vec{x} = 0$ defines a plane in $\mathbb{R}^3$ (not shown)
- Each such plane is orthogonal to its corresponding row vector $\vec{r}_i$
- The intersection of the two constraint planes is the purple line (the set of all $\vec{x}$ sent to $\vec{0}$ by B)

- Any $\vec{x}$ in this intersection is orthogonal to $\vec{r}_1$ and $\vec{r}_2$, hence orthogonal to every vector in row space(B)
- Dimension of null space(B) = n-r = 1 in $\mathbb{R}^3$



④ Codomain(B) is the $\mathbb{R}^2$ or generic plane

⑤ Matrix B takes a set of all vectors in $\mathbb{R}^3$ and transforms them in a way that

- (1, 0, 0) maps to $\vec{b}_1$
- (0, 1, 0) maps to $\vec{b}_2$
- (0, 0, 1) maps to $\vec{b}_3$
- all other vectors (x, y, z) follow
- B is a rank-2 matrix

↓

transformation by B is an onto transformation

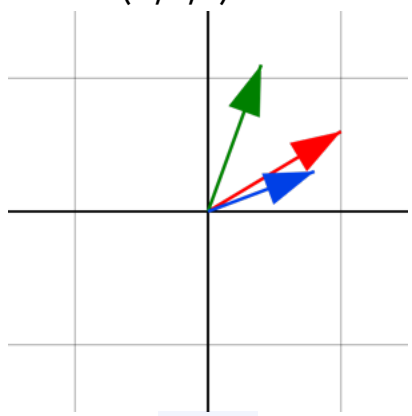
⑥ $\vec{r}_1$ & $\vec{r}_2$ act as 'sensors' that detect vectors that are not orthogonal to either

↓

purple line is orthogonal to both and is mapped to $\vec{0}$

Next image shows transformed basis vectors

- B (1,0,0) in red
- B (0,1,0) in green
- B (0,0,1) in blue



The four subspaces of A: summary table

- Domain ($\mathbb{R}^n$) = row space $\oplus$ null space

- Codomain ($\mathbb{R}^m$) = column space $\oplus$ ℓ-null space

Space	Lives in	Dim	Defined by	$\perp$ Complement	How it arises	Interpretation
null space(A)	$\mathbb{R}^n$	n−r	all $\vec{x}$ satisfying $A\vec{x} = \vec{0}$	row space(A)	Intersection of row constraints (in domain view)	Input directions that produce $\vec{0}$ output
row space(A)	$\mathbb{R}^n$	r	span{rows of A}	null space(A)	Spanned by independent rows (also $\perp$ to null(A))	Input directions that have no component in null(A)
col space(A)	$\mathbb{R}^m$	r	span{cols of A}	ℓ-null space(A)	Image of A in the codomain (span of columns)	Output directions that can be produced by A
ℓ-null space(A)	$\mathbb{R}^m$	m−r	all $\vec{y}$ satisfying $A^T\vec{y} = \vec{0}$ same as all $\vec{y}$ satisfying $\vec{y}^T A = \vec{0}^T$	col space(A)	Intersection of column constraints in the codomain	Output directions that have no component in column space

Note how the 'deduced' space orthogonal to null(A) becomes row space:

suppose

- $B = A^T$ and $B^T = A$
- row space(A) = col space(B) & col space(B) = row space(A)
- vector $\vec{y}$ satisfying $\vec{y}^T B = \vec{0}^T$ is in the ℓ-null space(B)
- same vector $\vec{y}$ also satisfies $B^T \vec{y} = \vec{0}$, same as $A \vec{y} = \vec{0}$
- same vector $\vec{y}$ belongs to null space(A)
- orthogonal complement of ℓ-null space(B) is column space(B), same as row space(A)

$\Downarrow$

The set of all vectors orthogonal to null space(A) is precisely the row space(A)

From the action of A, two spaces arise directly:

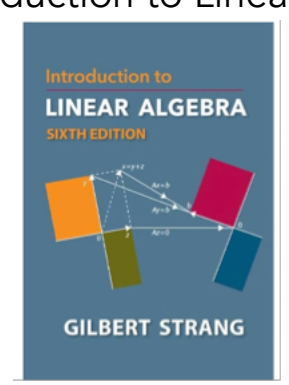
1. column space(A) in $\mathbb{R}^m$ (what A can produce, visualized in the codomain)
2. null space(A) in $\mathbb{R}^n$ (inputs sent to $\vec{0}$, visualized in the domain)

The remaining two subspaces are obtained as orthogonal complements of these

---

Inside the textbook image of the four subspaces

We now connect the same decomposition to the canonical image on the cover of Gilbert Strang, Introduction to Linear Algebra, MIT Press



Consider matrix A with dimensions (m=7)×(n=5) and rank r=3

① Image 1:

Domain(A) = $\mathbb{R}^5$ decomposes into

• row space(dim=r=3)

spanning directions: long blue arrows

• null space(dim=n-r=2)

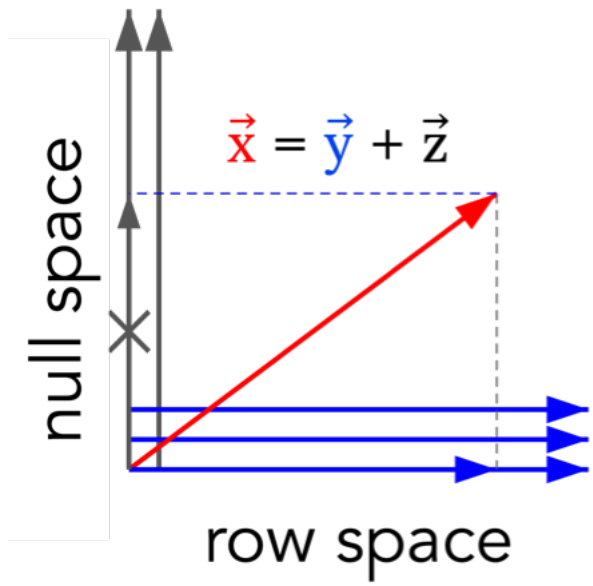
spanning directions: long gray arrows

• row space & null space intersect at $\vec{0}$

• every $\vec{y} \in$ row space(A) is orthogonal to every $\vec{z} \in$ null space

• every input vector $\vec{x} \in \mathbb{R}^5$ can be uniquely decomposed into

$\vec{y} \in$ row space(A) + $\vec{z} \in$ null space(A)



② Image 2:

Codomain(A) = $\mathbb{R}^7$ decomposes into

• col space(dim=r=3)

spanning directions: long blue arrows

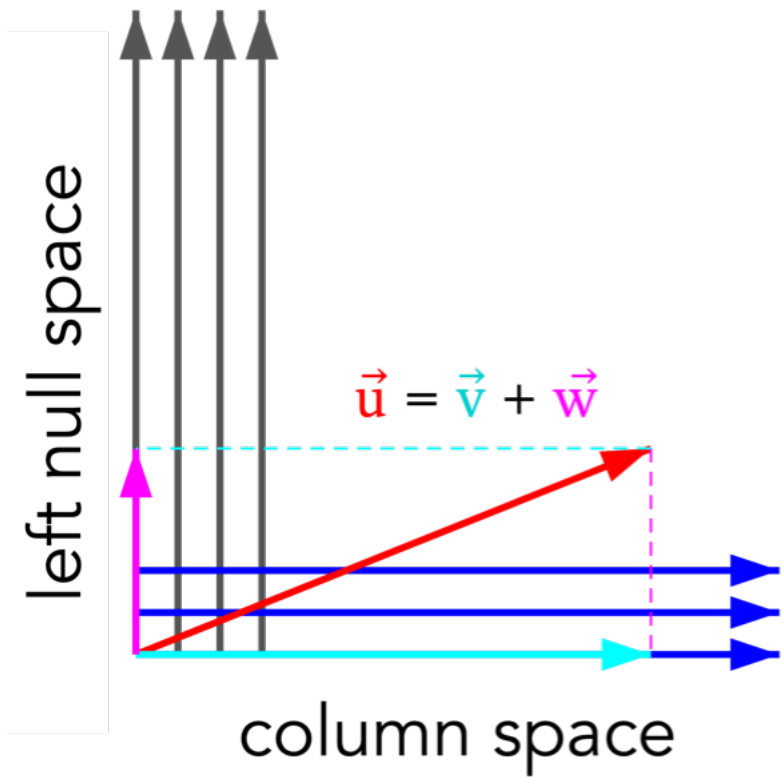
• ℓ-null space(dim=m-r=4)

spanning directions: long gray arrows

• col space & ℓ-null space intersect at $\vec{0}$

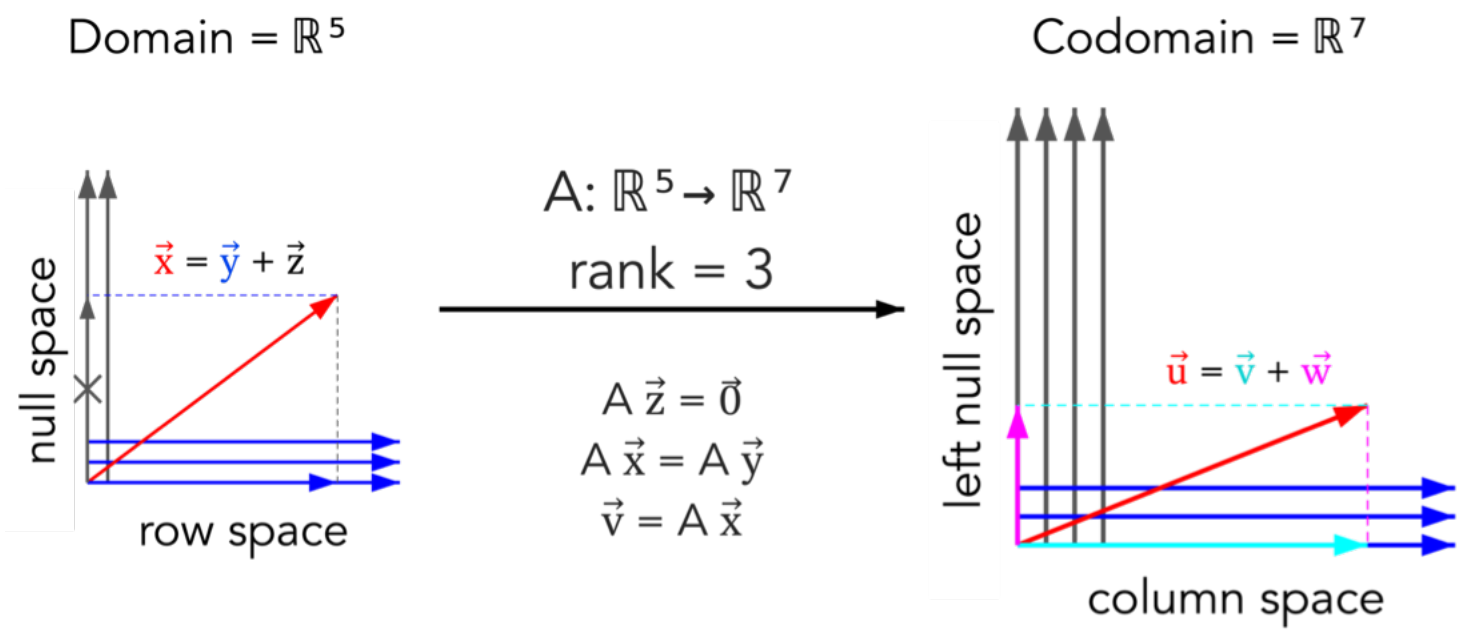
• every $\vec{v} \in$ col space(A) is orthogonal to every $\vec{w} \in \ell$-null space

• every output vector $\vec{u} \in \mathbb{R}^7$ can be uniquely decomposed into
$\vec{v} \in$ col space(A) + $\vec{w} \in \ell$-null space(A)



③ How A bridges input space to output space:
Applying A:
$A\,\vec{x} = A(\vec{y} + \vec{z}) = A\,\vec{y} + A\,\vec{z}$
since $\vec{z} \in$ null space(A), $A\,\vec{z} = \vec{0}$
$\Rightarrow A\,\vec{x} = A\,\vec{y}$

• the null-space component $\vec{z}$ disappears
• the output depends only on the row-space component $\vec{y}$
• all output vectors $A\,\vec{x}$ (= $A\,\vec{y}$) form col space(A)

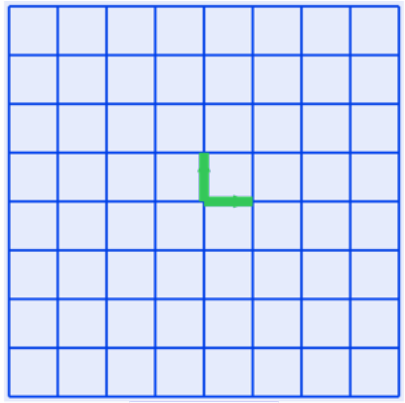


④ Orthogonal complement in the codomain:
ℓ-null space(A) is the set of all vectors $\vec{w}$ orthogonal to every vector $\vec{v} \in$ col space(A)
equivalently,
$\vec{w} \cdot \vec{v} = 0$ for all $\vec{v} \in$ col space(A)
or, in matrix form,
$\vec{w}^T A = 0$ or $A^T \vec{w} = 0$

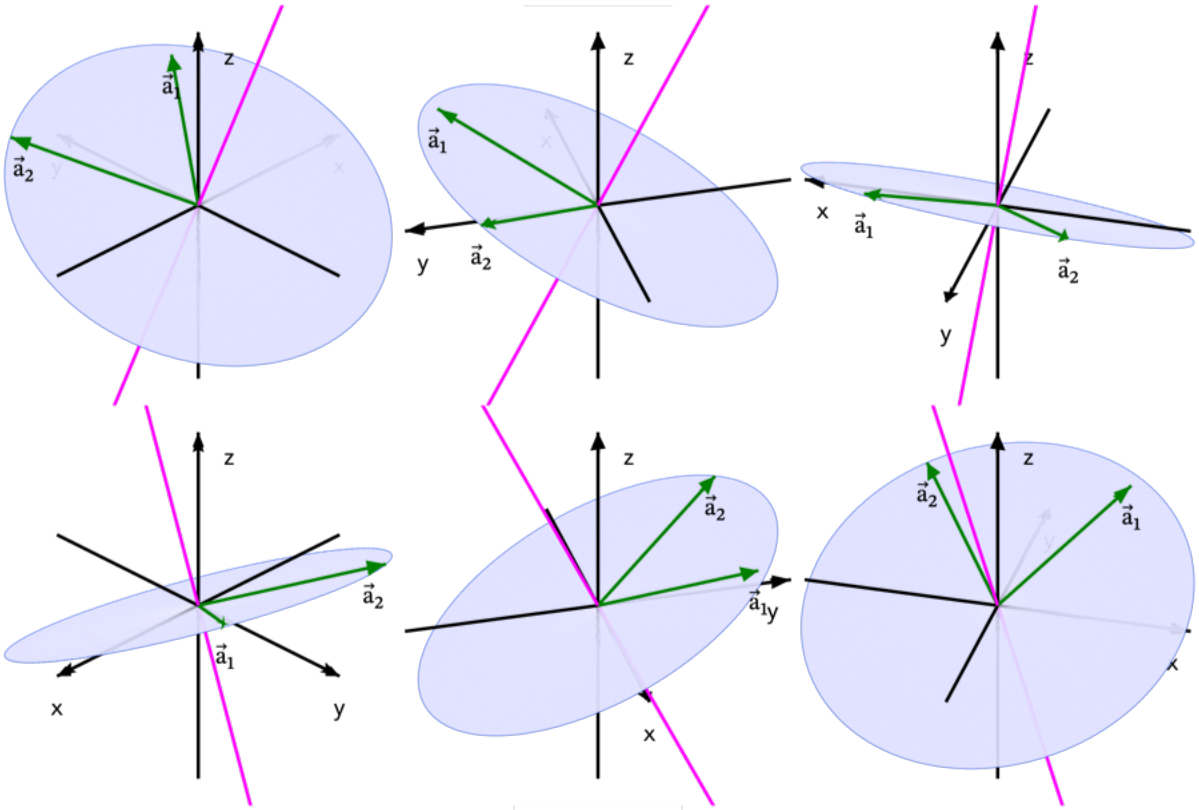
Geometric view of transformations by A, A^T & A^TA

We are looking at the same 3×2 matrix $A = [\, \vec{a}_1 \mid \vec{a}_2 \,]$
with linearly independent $\vec{a}_1$ and $\vec{a}_2$ in $\mathbb{R}^3$ and rank(A) = 2

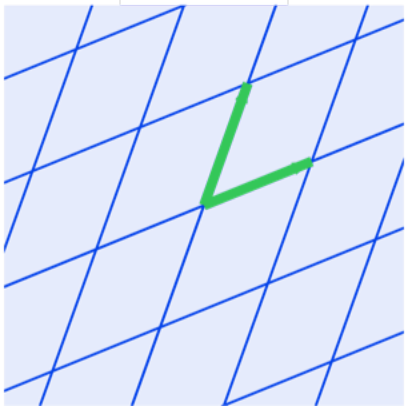
Transformations by A, followed by A^T



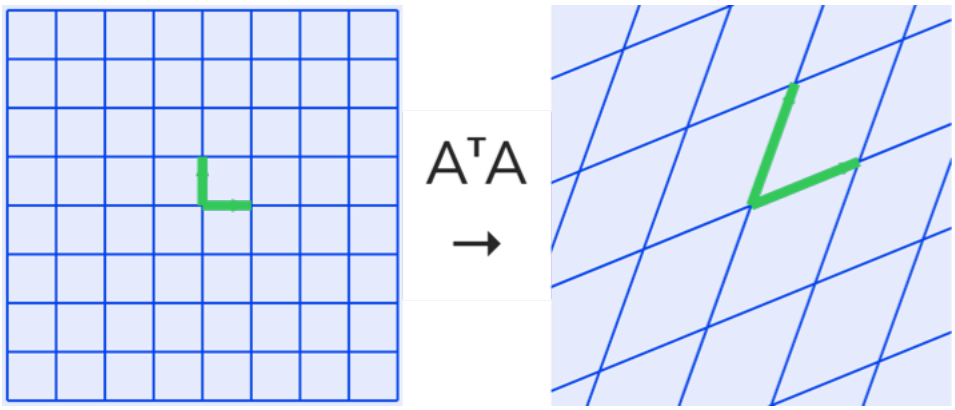
$A \Downarrow$



$A^T \Downarrow$



Direct transformation by $A^T A$



Note changing relationship between $\vec{a}_1$ & $\vec{a}_2$:

- They start as orthogonal
- After transformation by A, the grid loses squareness
- $A^T A$ brings this non-square grid back to $\mathbb{R}^2$ and

the final $A^T A$ grid looks more distorted than the grid seen in the embedded plane in $\mathbb{R}^3$

This is because $A^T A$ applies the same distortion twice:
once to embed the grid via A and once pull that geometry back via A^T

Suppose $\vec{a}_1$ & $\vec{a}_2$ are orthogonal with length = 1:
In this case $A^T A$ is identity and the grid returns to original state

This is how to measure 'grid distortion' by A:
Suppose we have vectors $\vec{x}$ & $\vec{y}$ in the original plane, so their dot product = $\vec{x}^T \cdot \vec{y}$
After transformation by A, dot product becomes

$$(A\,\vec{x})^T \cdot (A\,\vec{y}) = \vec{x}^T A^T A \cdot \vec{y} = \vec{x}^T \cdot (A^T A\,\vec{y})$$

The four subspaces of A^T

We now apply the same construction to the transpose matrix A^T:
rows of A^T are columns of A, and columns of A^T are rows of A

Space	Lives in	Dim	Spanned by	⊥ Complement
row space(A^T)	$\mathbb{R}^m$	r	Rows of A^T (Cols of A)	null space(A^T)
null space(A^T)	$\mathbb{R}^m$	m−r	all $\vec{x}$ satisfying $A^T \vec{x} = 0$	row space(A^T)
col space(A^T)	$\mathbb{R}^n$	r	Cols of A^T (Rows of A)	ℓ-null space(A^T)
ℓ-null space(A^T)	$\mathbb{R}^n$	n−r	all $\vec{b}$ satisfying $A \vec{b} = 0$	col space(A^T)

Connecting the null spaces of A and A^T

① null(A) = ℓ-null(A^T)

Proof:
Let $\vec{x}$ be a vector in $\mathbb{R}^n$

Recall the transpose identity for matrix-vector multiplication:
$(A \vec{x})^T = \vec{x}^T A^T$

Now prove set inclusion in both directions

- If $\vec{x} \in$ null(A), then $A \vec{x} = 0$
Taking transpose gives $(A \vec{x})^T = 0^T = 0$
Using the identity above, $\vec{x}^T A^T = 0$
So $\vec{x} \in$ ℓ-null(A^T)

‣ If $\vec{x} \in \ell\text{-null}(A^T)$, then $\vec{x}^T A^T = 0$
Taking transpose gives $(\vec{x}^T A^T)^T = 0$
But $(\vec{x}^T A^T)^T = A\vec{x}$
So $A\vec{x} = 0$, hence $\vec{x} \in \text{null}(A)$

Since each set is contained in the other, the two sets are equal

② $\ell\text{-null}(A) = \text{null}(A^T)$

Proof:
Let $\vec{y}$ be a vector in $\mathbb{R}^m$

Recall the transpose identity:
$(\vec{y}^T A)^T = A^T \vec{y}$

‣ If $\vec{y} \in \ell\text{-null}(A)$, then $\vec{y}^T A = 0$
Taking transpose gives $A^T \vec{y} = 0$
So $\vec{y} \in \text{null}(A^T)$

‣ If $\vec{y} \in \text{null}(A^T)$, then $A^T \vec{y} = 0$
Taking transpose gives $\vec{y}^T A = 0$
So $\vec{y} \in \ell\text{-null}(A)$

Thus the two spaces are also equal

The four subspaces of A^TA

Consider any $m \times n$ matrix A

- $A^T A$ is $n \times n$ matrix, which maps $\mathbb{R}^n$ to $\mathbb{R}^n$

$\Leftrightarrow$

Domain of $A^T A$ is identical to its codomain

- $A^T A$ is symmetric: $(A^T A)^T = A^T (A^T)^T = A^T A$

① $\text{null}(A^T A) = \text{null}(A)$

Suppose vector $\vec{x} \in$ null space(A)

or

$A\vec{x} = \vec{0}$

Then

$(A^T A)\vec{x} = A^T(A\vec{x}) = \vec{0}$

$\Downarrow$

null space$(A) \subseteq$ null space$(A^T A)$ (is a subset of)

To prove the reverse inclusion,

suppose vector $\vec{y} \in$ null space$(A^T A)$

or

$(A^T A)\vec{y} = \vec{0}$

Multiply on the left by $\vec{y}^T$:

$\vec{y}^T (A^T A)\vec{y} = 0$

$\vec{y}^T (A^T A)\vec{y} = (A\vec{y})^T (A\vec{y}) = \|A\vec{y}\|^2$

So $\|A\vec{y}\|^2 = 0 \Rightarrow A\vec{y} = \vec{0}$

$\Downarrow$

null space$(A^T A) \subseteq$ null(A)

Together:

null space$(A^T A) =$ null space(A)

② $A^T A$ is symmetric

$\Downarrow$

- row space$(A^T A) =$ column space$(A^T A)$
- ℓ-null space$(A^T A) =$ null space$(A^T A)$

$\Downarrow$

four subspaces of A reduce to two

Space	Lives in	Dim	Spanned by	$\perp$ Complement
row space($A^T A$) = col space($A^T A$)	$\mathbb{R}^n$	r	rows (same as cols) of $A^T A$	null space($A^T A$) = ℓ-null space($A^T A$)
null space($A^T A$) = ℓ-null space($A^T A$) = null space(A)	$\mathbb{R}^n$	n−r	all $\vec{x}$ satisfying $(A^T A)\,\vec{x} = \vec{0}$ or $\vec{x}^T (A^T A) = \vec{0}^T$	row space($A^T A$) = col space($A^T A$)

Suppose A is a 'tall' matrix with m>n and full column rank

$\Downarrow$

$A^T A$ has full rank = n

$\Downarrow$

$(A^T A)\,\vec{x} = A^T \vec{b}$ has a single solution

Importance of this concept will be illustrated on the next page

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Why do we need this abstract concept?

Suppose we have an inconsistent system of equations $A\,\vec{x} = \vec{b}$ where

- $\vec{b}$ cannot be presented as a linear combination of columns of A

$\Leftrightarrow$

there is no $\vec{x}$ that satisfies the equation

- A has full column rank $\Leftrightarrow$ $(A^T A)^{-1}$ is defined

So we set the goal of finding the closest $\vec{x}$

Recall from the page on the four subspaces that every $\vec{b}$ in $\mathbb{R}^m$ (or co-domain of A)

can be written as a sum of $\left(\vec{b}_{\parallel} \in \text{col}(A)\right) + \left(\vec{b}_{\perp} \in \ell\text{-null}(A)\right)$

or

$$\vec{b} = \vec{b}_{\parallel} + \vec{b}_{\perp}$$

Also recall from previous page that ℓ-null(A) = null(A^T)

$$\Downarrow$$

$$\vec{b}_{\perp} \in \text{null}(A^T)$$

or, by definition,

$$A^T \vec{b}_{\perp} = \vec{0}$$

To define the goal of finding the closest solution,
we want to solve for such vector that

- belongs to col(A)
- is closest to $\vec{b}$

The vector we are looking for happens to be $\vec{b}_{\parallel}$,
which takes us to solving

$$A\vec{x} = \vec{b}_{\parallel}$$

or, equivalently,

$$A\vec{x} = \vec{b} - \vec{b}_{\perp}$$

We can multiply both sides by A^T:

$$A^T A\vec{x} = A^T(\vec{b} - \vec{b}_{\perp}) = A^T\vec{b} - A^T\vec{b}_{\perp} = A^T\vec{b} - \vec{0}$$

A has full column rank, therefore $(A^T A)^{-1}$ is defined, and
we can multiply both sides by $(A^T A)^{-1}$:

$$(A^T A)^{-1} A^T A\vec{x} = (A^T A)^{-1} A^T \vec{b}$$

or

$$\vec{x} = (A^T A)^{-1} A^T \vec{b}$$

Image for (m=3)×(n=2) matrix A with linearly independent columns: r=2
The blue plane is col(A) = span($\vec{a}_1$, $\vec{a}_2$)
The 'unreachable' red vector $\vec{b}$ is outside span of the columns

We decompose $\vec{b}$ as $\vec{b}$|| (teal) + $\vec{b}\perp$ (magenta)

- $\vec{b}\| \in$ col(A)
- $\vec{b}\perp \perp$ col(A)

So $\vec{b}$|| is the closest vector to $\vec{b}$ that lies in col(A)
and the least squares condition is $A\,\vec{x} = \vec{b}\|$

Note that the image shows the following structures:

1. Entire $\mathbb{R}^3$
2. Plane as subspace of $\mathbb{R}^3$, or $\mathbb{R}^2$ space in its own right with its own coordinate system

- Matrix A inputs vectors represented in coordinates of $\mathbb{R}^2$
and represents them in coordinates of $\mathbb{R}^3$

$\Leftrightarrow$

maps $\mathbb{R}^2 \to \mathbb{R}^3$

- Matrix A^T inputs vectors represented in coordinates of $\mathbb{R}^3$
and represents them in coordinates of $\mathbb{R}^2$

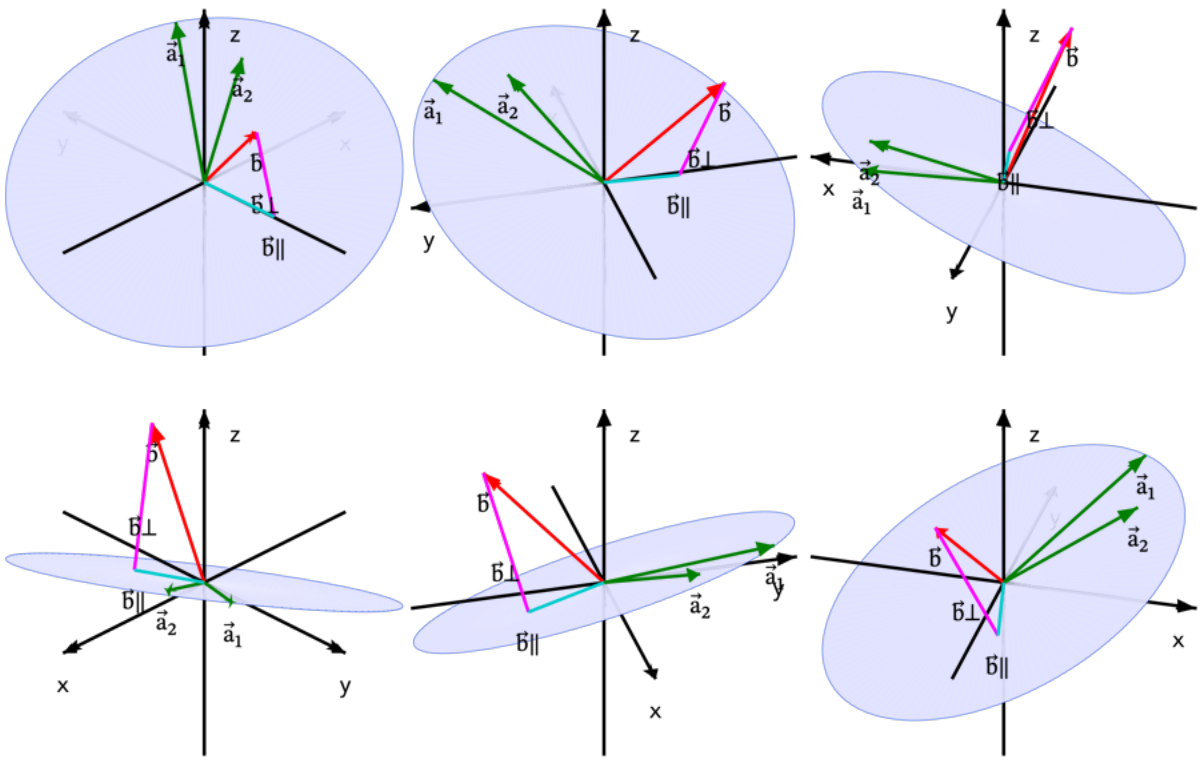
$\Leftrightarrow$

maps $\mathbb{R}^3 \to \mathbb{R}^2$

$\Downarrow$

left-multiplying by A^T takes $\vec{b}$ and outputs a 2D vector
whose entries are dot products with directions $\vec{a}_1$ and $\vec{a}_2$
These numbers describe how $\vec{b}$|| aligns with the spanning directions of the plane
They are not yet the expansion coefficients:
The expansion coefficients $\vec{x}$ are obtained by solving $A^T A\,\vec{x} = A^T\vec{b}$ as demonstrated

above



Tying it all together

Condition	Equivalent to
null space = $\{\vec{0}\}$	• $A\vec{x} = \vec{0}$ has only trivial solution • no free variables • if $A\vec{x}1 = A\vec{x}2$ then $\vec{x}1 = \vec{x}2$ • transformation is one-to-one
null space ≠ $\{\vec{0}\}$	• $A\vec{x} = \vec{0}$ has nontrivial solutions • free variables exist • if $A\vec{x}1 = A\vec{x}2$ then $\vec{x}1$ and $\vec{x}2$ may be different • transformation is not one-to-one
ℓ-null space = $\{\vec{0}\}$	• column space = codomain • every $\vec{b}$ in codomain has a solution to $A\vec{x} = \vec{b}$ • no impossible outputs • transformation is onto
ℓ-null space ≠ $\{\vec{0}\}$	• column space ≠ codomain • some vectors $\vec{b}$ in the codomain have no solution to $A\vec{x} = \vec{b}$ • outputs with nonzero ℓ-null component cannot be produced • transformation is not onto